Spectral diagnostics and local fixed-budget sensitivity of critical points in finite encounter-reaction models

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Reaction-time densities of confined encounter reactions can contain several transport-generated time windows, yet whether those windows appear as critical points of the measured flux depends jointly on transport, the initial law, and the spatial reaction field. We develop an exact finite-model calculus for diagnosing and steering those critical points. First, a reaction-support Green reduction separates free transport from volume killing. Second, the generalized Descartes rule gives a necessary spectral gate: a reversible finite-model density with m interior modes requires at least $2m-1$ sign changes in its ordered nonzero residues. Third, an exact Fréchet–Duhamel derivative gives the response of the critical-point equation to every killing rate, and projection onto the tangent space of a declared reactivity budget identifies the locally steepest feasible redistribution in a chosen metric.

Two families of finite generators exercise the calculus. Direct matrix-exponential derivatives locate a nondegenerate reaction-density fold in a finite two-reactant continuous-time Markov chain under a rate-ratio control, and independent continuation recovers the generic 1/2 critical-point-separation and 3/2 prominence exponents of the fold normal form. A separate finite-radius two-dimensional family redistributes reactivity at an exactly fixed per-grid killing sum and exhibits the same nondegenerate fold structure and exponents on three grids. The critical controls of the two-dimensional family are grid- and budget-measure sensitive, so they are reported as finite-generator mechanism certificates rather than continuum critical values, and the convergence calculation that a continuum bridge would require is stated explicitly. Detailed endpoint, coordinate, clock, capacity, and multichannel audits are reported in the Supplemental Material.

I. INTRODUCTION

The reaction time of two mobile particles can retain information that a mean reaction time or a single long-time decay rate suppresses. In confinement, first contact need not be reaction: encounter can be imperfect and reaction can be restricted to selected spatial regions. The measured density then depends jointly on transport, encounter geometry, and the reaction observable. Multiple critical points of the density—time-stationary points in the sense of singularity theory, not critical phenomena—can reflect dynamically distinct reaction streams, but their existence does not by itself identify which mechanism produced them.

Multimodal passage and reaction-time laws have established transport, geometric, internal-state, and encounter-history mechanisms. Confined encounter laws, exact propagators, heterogeneous transmission sites, multi-target reductions, and transport-generated multiple peaks are developed in Refs. [1–9]. Imperfect reaction has likewise been treated through Doi volume killing, radiation boundaries, and heterogeneous reactivity [10–15]. Static killing fields and partially reactive network elements have already been rearranged at fixed total strength, and position-dependent Robin reactivity has been optimized under an integral constraint [16–18]. Besga *et al.* previously established a Brownian first-passage shape transition in which a finite-time maximum–minimum pair merges under a target-distance/initial-spread control [19]. We therefore do not claim either the fold criterion $f_t = f_{tt} = 0$, a two-clock shape transition, or any of the other ingredients above as new.

The narrower problem here is differential. Suppose a finite killed generator has a reaction-density fold, so that $f_t = f_{tt} = 0$ at positive time. Can one (i) determine whether its observable spectrum has enough sign variation to permit the requested number of modes, (ii) compute the response of the fold equation to a spatial killing redistribution, and (iii) identify the most effective infinitesimal redistribution compatible with a fixed budget? We answer those questions exactly for declared finite generators, then test the resulting calculus in a finite encounter chain and a finite-radius two-dimensional family.

The distinction between a *strict extremum* and a classifier-*resolved mode* is important. The main article uses derivative-certified folds only. Endpoint morphology, synchronous-clock controls, catalytic-coordinate sensitivity, finite-grid trimodality, capacity calibration, and generalized-inverse-Gaussian (GIG) screening are preserved as secondary audits in the standalone Supplemental Material [20]. Those calculations delimit the causal language but are not needed to establish the two fold certificates presented here.

The five main claims and their boundaries are summarized in Table I. Figure 1 shows the spatial channel

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picture and the generic fold geometry.

II. FINITE ENCOUNTER MODEL AND GREEN REDUCTION

Let $X_i(t) \in \Omega \subset \mathbb{R}^d$ be two independent particles before reaction, with generator $\mathcal{L}_0$ and reflecting outer boundaries. A finite contact radius $a > 0$ and an affine catalytic coordinate

$$C_\eta(x_1, x_2) = \eta x_1 + (1-\eta)x_2, \qquad 0 \le \eta \le 1,$$

define channel killing fields

$$\begin{aligned} K_{a,j}^{(\eta)} &= \kappa_j(C_\eta)\mathbf{1}_{C_j}(C_\eta)\mathbf{1}_{\{|x_1-x_2|<a\}}, \\ K_a^{(\eta)} &= \sum_j K_{a,j}^{(\eta)}. \end{aligned} \tag{1}$$

The unreacted joint density obeys

$$\partial_t p = \mathcal{L}_0 p - K_a^{(\eta)} p. \tag{2}$$

With $S(t) = \int p$ and $f_j(t) = \int K_{a,j}^{(\eta)} p$, reflecting boundaries give the exact mass balance

$$-S'(t) = f(t) = \sum_j f_j(t). \tag{3}$$

The catalyst coordinate is part of the physical model: the midpoint and the diffusion-decoupling weighted centre are unequal when the particle diffusivities differ. Coordinate algebra, the transformed domain, Doi–Robin qualifications, and the finite-grid midpoint audit are given in the Supplemental Material [20].

Write the volume killing operator on $\mathsf{H} = L^2(\mathcal{D})$ as

$$V = \Gamma^* \mathsf{K} \Gamma, \qquad \mathcal{T} = \mathcal{L}_0 - V, \tag{4}$$

where Γ restricts a function to the reaction support, Γ^* extends it by zero, and K is a bounded nonnegative multiplication operator on that support. It may have a kernel. For $\operatorname{Re} s$ above the spectral bound, define

$$\mathcal{R}_0(s) = (s - \mathcal{L}_0)^{-1}, \qquad \mathcal{G}(s) = \Gamma \mathcal{R}_0(s) \Gamma^*. \tag{5}$$

Whenever the displayed inverse exists, the inverse-free resolvent identity is

$$(s - \mathcal{T})^{-1} = \mathcal{R}_0 - \mathcal{R}_0 \Gamma^* \mathsf{K}(I + \mathcal{G}\mathsf{K})^{-1} \Gamma \mathcal{R}_0. \tag{6}$$

Thus, for $u = \Gamma \mathcal{R}_0 q_0$, the restricted state and transformed reaction density are

$$x = (I + \mathcal{G}\mathsf{K})^{-1} u, \qquad y = \mathsf{K}(I + \mathcal{G}\mathsf{K})^{-1} u. \tag{7}$$

No inverse of K is required. This separation exposes free transport through $\mathcal{G}$, geometry through Γ, and intrinsic chemistry through K.

For a finite continuous-time Markov chain (CTMC), adopt row vectors. Let $L_0 = L_1 \oplus L_2$, let columns of U select reactive states, and let K be diagonal in channel rates. Then

$$T = L_0 - UKU^{\mathsf{T}}, \qquad f_j(t) = \alpha \mathrm{e}^{Tt} UKe_j. \tag{8}$$

The finite support matrix

$$G(s) = U^{\mathsf{T}}(sI - L_0)^{-1} U$$

gives

$$\det(sI - T) = \det(sI - L_0) \det[I + G(s)K].$$

The second determinant locates only candidate killed poles coupled to the reaction support. Dark transport modes, numerator cancellation, and observable residues must still be checked. Splitting probabilities are computed from the stable killed solve

$$\boldsymbol{\pi} = \alpha(-T)^{-1} UK, \tag{9}$$

when the reachable live chain is transient. Exact time derivatives are

$$\partial_t^n f(t) = \alpha \mathrm{e}^{Tt} T^n UK\mathbf{1}. \tag{10}$$

The detailed two-site exclusion test model and zero-frequency diagnostics are reported in the Supplemental Material [20].

III. SPECTRAL GATE AND FIXED-BUDGET SUSCEPTIBILITY

A. A necessary sign-variation condition

Consider a finite killed model whose observable density has a real, diagonalizable spectral expansion. After equal decay rates are grouped and zero residues removed, write

$$f(t) = \sum_{j=1}^{n} a_j \mathrm{e}^{-\lambda_j t}, \qquad 0 < \lambda_1 < \cdots < \lambda_n. \tag{11}$$

Reversibility is a sufficient hypothesis because detailed balance makes the killed generator similar to a real symmetric matrix. Applying the classical generalized Descartes rule [21] to

$$f_t(t) = -\sum_{j=1}^{n} \lambda_j a_j \mathrm{e}^{-\lambda_j t} \tag{12}$$

after $x = \mathrm{e}^{-t}$ gives the following corollary:

$$\#\{\text{positive-time zeros of } f_t\} \le V(a_1, \ldots, a_n),$$

where zeros are counted with multiplicity and V is the number of ordered residue sign changes. Hence m nondegenerate interior maxima separated by minima require

$$V(a_1, \ldots, a_n) \ge 2m - 1. \tag{13}$$

TABLE I. Evidence hierarchy used in the main article.

Statement	Evidence	Boundary
Reaction-support Green reduction	Exact conditional operator identity and finite Woodbury specialization	Volume killing on a declared support; no unqualified Robin trace identity or continuum meromorphic continuation
Finite reversible spectral gate	Generalized-Descartes corollary and primary-generator reconstruction	Necessary, not sufficient; not a channel-rank bound and not applied to complex-spectrum, Jordan, or continuum laws
Fixed-budget modality susceptibility	Exact finite Fréchet–Duhamel identity, projected optimum, and independent directional checks	Infinitesimal interior-rate design in a declared budget and metric; no finite-amplitude, binary-mask, moving-support, or continuum optimum
Finite encounter-chain fold	Direct finite-matrix derivatives, root solve, residuals, transversality, and independent normal-form scaling	One finite model; minority-channel splitting probability 8.95×10^{-6}
Finite-radius 2D matched-budget folds	Three finite-generator folds, independent normal-form scaling, and a second budget-measure audit	Critical controls are nonmonotone and budget-sensitive; no converged continuum critical control

This is a necessary falsification gate, not a mode-count predictor. Three sign changes can coexist with only one positive critical point, and a rank-one killing support can already produce a bimodal phase-type density. Reaction-support rank therefore cannot replace the observable residues.

Full symmetric eigendecompositions rebuild the two primary generators used in the numerical audits. Reversibility, the gate's hypothesis, is verified directly for both free generators: every live transition is two-way, a spanning-tree reconstruction of the detailed-balance log-weights closes on all off-tree edges with maximum cycle inconsistency 2.4×10^{-14} (961 states) and 9.1×10^{-14} (2025 states), and the recovered measure symmetrizes the killed generator. Equal decay rates are grouped at a relative tolerance 2×10^{-10} before residues are summed; at that tolerance no two eigenvalues of either generator merge. The 961-state finite encounter generator is evaluated at its three reported critical points; the separate 2025-state three-channel generator is evaluated at five detected critical points. Across the four relative residue cutoffs 10^{-12}, 10^{-10}, 10^{-8}, and 10^{-6} (relative to the summed absolute grouped residues), the minimum retained sign counts are 444 and 191, respectively, above the necessary bounds three and five. Because these reconstructions reuse the same generator constructors and critical-time artifacts, they test decomposition and reconstruction rather than discover roots independently. Full cutoff, counterexample, and reconstruction details are in the Supplemental Material [20].

B. Budget-projected modality response

For a finite live-state transport generator L, let

$$T(k) = L - \operatorname{diag} k, \qquad f(t;k) = \alpha \mathrm{e}^{T(k)t} k, \tag{14}$$

where $k > 0$ on the allowed design support. For $k_\epsilon = k + \epsilon h$, put $H = -\operatorname{diag} h$. The exact Fréchet–Duhamel identity is

$$D\mathrm{e}^{Tt}[H] = \int_0^t \mathrm{e}^{T(t-s)} H \mathrm{e}^{Ts}\, \mathrm{d}s \tag{15}$$

[22]. Since $f_t = \alpha \mathrm{e}^{Tt} Tk$, its complete directional derivative is

$$D_k f_t(t;k)[h] = \alpha D\mathrm{e}^{Tt}[H]Tk + \alpha \mathrm{e}^{Tt}(Hk + Th). \tag{16}$$

The second term is essential because the killing field is also the measured reaction observable. Writing this linear functional as $g^\mathsf{T} h$,

$$g_i = (\alpha \mathrm{e}^{Tt} T)_i - k_i(\alpha \mathrm{e}^{Tt})_i - \int_0^t (\alpha \mathrm{e}^{T(t-s)})_i (\mathrm{e}^{Ts} Tk)_i\, \mathrm{d}s. \tag{17}$$

Let $c > 0$ define the declared budget $c^\mathsf{T} h = 0$, and let $M \succ 0$ define the local norm $h^\mathsf{T} M h = 1$. Put

$$\lambda_B = \frac{c^\mathsf{T} M^{-1} g}{c^\mathsf{T} M^{-1} c}, \qquad \widetilde{g} = g - \lambda_B c. \tag{18}$$

Cauchy–Schwarz on the budget tangent space gives

$$h_* = \frac{M^{-1}\widetilde{g}}{\sqrt{\widetilde{g}^\mathsf{T} M^{-1} \widetilde{g}}}, \qquad \max D_k f_t[h] = \sqrt{\widetilde{g}^\mathsf{T} M^{-1} \widetilde{g}}. \tag{19}$$

At a double stationary point, $\widetilde{g} \neq 0$ means that some infinitesimal feasible redistribution is transverse to the fold. The direction depends on the selected budget c and metric M; it is not a coordinate-free global optimum.

An eleven-state test model, with seeded random positive diagonal metric M and budget vector c (uniform draws on $[0.65, 1.85]$ and $[0.55, 1.65]$), compares Eq. (17) with a basis Fréchet calculation, 256 independently generated budget-zero finite differences, a state permutation, and 20,000 random feasible directions. The relative

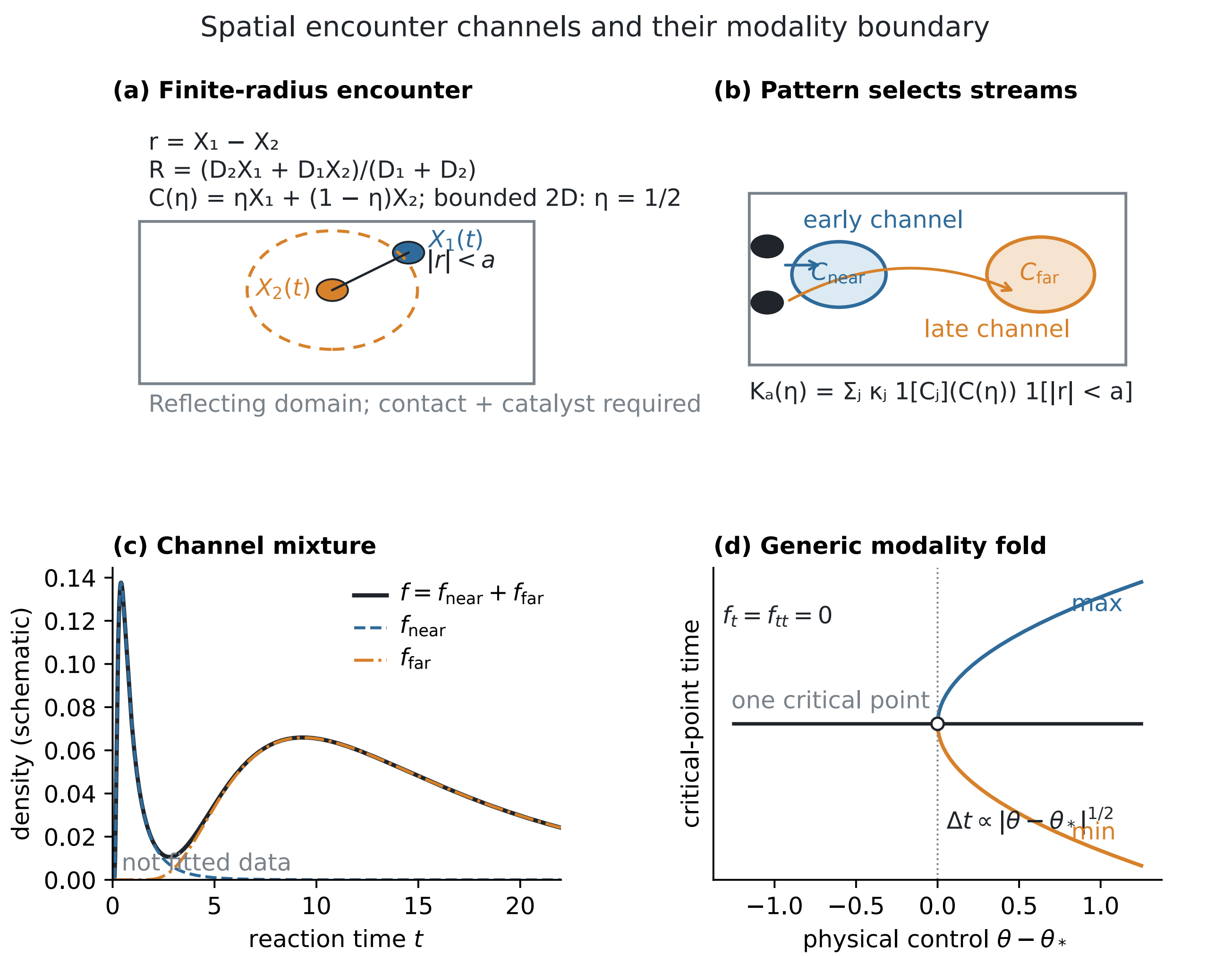

FIG. 1. Spatial encounter channels and their modality boundary. (a) Reaction requires finite-radius contact and catalytic activation at a declared affine encounter location. (b) Separated patches select early and late streams. (c) Their sum can be bimodal; the curves are schematic and are not fitted data. (d) At a generic fold, a maximum–minimum pair is created with square-root separation under a physical control.

Duhamel error is 3.3×10^{-15}, the maximum independent finite-difference error is 2.4×10^{-8}, and the best random response is 0.933 of Eq. (19). At the finite encounter fold below, the chain-rule reconstruction of the computed log-rate transversality agrees to 6.8×10^{-15} relative error. The orthogonal near-to-far fixed-state-sum redistribution has nonzero transversality -1.1621×10^{-5}.

IV. PHYSICAL FOLDS IN FINITE ENCOUNTER GENERATORS

At a fold of density critical points under a physical parameter θ,

$$f_t(t_*,\theta_*) = f_{tt}(t_*,\theta_*) = 0. \tag{20}$$

The generic nondegeneracy conditions are

$$a_* = f_{t\theta}(t_*,\theta_*) \neq 0, \qquad b_* = f_{ttt}(t_*,\theta_*) \neq 0. \tag{21}$$

Taylor expansion gives

$$f_t = a_*\Delta\theta + \frac{b_*}{2}\Delta t^2 + o(|\Delta\theta| + |\Delta t|^2).$$

On the side where $-2a_*\Delta\theta/b_* > 0$, the new maximum and minimum separate as

$$\Delta t_\pm = \pm\sqrt{-\frac{2a_*}{b_*}\Delta\theta} + o(|\Delta\theta|^{1/2}), \tag{22}$$

and integrating the quadratic derivative gives prominence proportional to $|\Delta\theta|^{3/2}$. These equations concern a physical control that can change both channel weights and shapes; a freely adjusted mixture weight is not substituted for θ.

A. Finite two-reactant CTMC certificate

The finite encounter chain has sites $0,\dots,30$. Its two independently clocked walkers have interior total (attempt) jump rates 0.7 and 0.2, respectively, and a common right bias 0.3. Their interior left/right rates are therefore $(0.245, 0.455)$ and $(0.07, 0.13)$. At a boundary an out-of-range jump is omitted, giving the declared reflecting (holding) convention. If L_1 and L_2 are the corresponding row generators, the free pair generator is the Kronecker sum $L_1 \oplus L_2 = L_1 \otimes I + I \otimes L_2$, and the initial law is concentrated at $(3, 15)$. Killing acts only at the reactive co-locations $(13, 13)$ and $(30, 30)$, with rates κ_{near} and κ_{far}, respectively. Transport and the initial law remain fixed. We set $\kappa_{\mathrm{far}} = -\log 0.4 = 0.9162907319$ and vary

$$\theta = \log(\kappa_{\mathrm{near}}/\kappa_{\mathrm{far}}), \tag{23}$$

so that $\kappa_{\mathrm{near}} = \kappa_{\mathrm{far}} e^{\theta}$. Parameter derivatives are computed by propagating the column state and its sensitivity,

$$p' = T^{\mathsf{T}} p, \qquad s' = T^{\mathsf{T}} s + T_\theta^{\mathsf{T}} p, \qquad s(0) = 0, \tag{24}$$

in one augmented matrix exponential; all semigroup actions are sparse matrix-exponential actions in IEEE double precision. Time derivatives use Eq. (10); no sampled-curve maximum is used to locate the fold, which is obtained by a Powell hybrid root solve on the scaled residual pair (f_t, f_{tt}) with parameter tolerance 10^{-11} and guarded acceptance thresholds.

The numerical root is

$$t_* = 37.0749586402, \qquad \theta_* = -9.6753635853, \tag{25}$$

corresponding to $\kappa_{\mathrm{near}} = 5.755410148 \times 10^{-5}$. Residuals and margins are made dimensionless with the fold time and density: $|f_t| t_*/f = 7.1 \times 10^{-15}$, $|f_{tt}| t_*^2/f = 6.8 \times 10^{-13}$, third derivative $f_{ttt} t_*^3/f = 69.94$, and parameter transversality $f_{t\theta} t_*/f = -0.1962$. Independent continuation uses a symmetric geometric ladder of fourteen control offsets $\Delta\theta = \pm 0.001$–0.064 that play no role in locating the fold; the log–log fits use the six two-peak-side offsets $\Delta\theta \le 0.032$ and give slopes 0.50095 for separation and 1.50877 for prominence, while the seven one-peak-side offsets confirm the absence of any critical pair (Fig. 2). The normal-form amplitude is tested as well: the predicted separation $2\sqrt{-2a_*/b_*}\,\Delta\theta^{1/2} = 5.554\,\Delta\theta^{1/2}$ matches the measured separations to 0.01% at $\Delta\theta = 0.001$ and 0.7% at $\Delta\theta = 0.064$. Mass closure is 4.0×10^{-15}, and survival at $t = 5000$ is 1.3×10^{-21}.

The very small near-channel probability is reported as a limitation. The fold is derivative-certified even though the emerging feature is not a practically balanced two-peak signal. The associated synchronous discovery, Poissonized-clock comparison, and GIG approximation errors are secondary mechanism checks and are kept in the Supplemental Material [20].

B. Finite-radius two-dimensional matched-budget folds

The second family—labeled M2D-F, for matched-budget two-dimensional finite-radius—is a separate bounded realization. Two particles evolve on the obstacle-free unit square with reflecting outer boundaries. Each one-particle generator is a conservative boundary-node CTMC with central nearest-neighbor diffusion, upwind longitudinal drift, and a smooth transverse restoring drift: on the uniform grid with spacings h_x, h_y, a node jumps to its right or left neighbor at rate $D/h_x^2 + \max(\pm v_x, 0)/h_x$ and to its upper or lower neighbor at rate $D/h_y^2 + \max(\pm v_y, 0)/h_y$, with drift field $(v_x, v_y) = (v, -\gamma_\perp(y - \frac{1}{2}))$, and out-of-range jumps are omitted (reflection by omission). Reaction occurs when the particle separation is at most $a = 0.17$ and the midpoint $C_{1/2} = (X_1 + X_2)/2$ lies in one of two circular catalytic patches. The parameters are

$$(D_1, v_1) = (0.0025, 0.115), \qquad (D_2, v_2) = (0.0008, 0.02),$$

with transverse confinement 1.5, starts $(0, 0.5)$ and $(0.28, 0.5)$, and near/far patch specifications

$$(x, r, \kappa)_{\mathrm{near}} = (0.25, 0.18, 0.5),$$
$$(x, r, \kappa)_{\mathrm{far}} = (0.72, 0.20, 15),$$

where both patch centers have $y = 0.5$. The full closed-mask convention, contact-safe initial-law construction, and dimensional scaling are specified in the Supplemental Material [20]. Because the tested grids have large local Péclet numbers and binary supports, they are treated as finite generators rather than a controlled SDE refinement.

On each grid, a homogeneous rate $\bar\kappa_h$ is recomputed so that the unweighted statewise killing sum equals that of the patterned endpoint. The reactive-tube state counts and matched rates are $(N_T, N_{\mathrm{near}}, N_{\mathrm{far}}, \bar\kappa_h) = (125, 7, 9, 1.108)$, $(349, 34, 40, 1.768)$, and $(1123, 109, 137, 1.878)$ on the 9×5, 11×7, and 13×9 grids, and the build aborts unless the killing sum at both path endpoints matches the budget to 2×10^{-12}. The physical path is

$$\kappa_\theta(C_{1/2}) = (1-\theta)\bar\kappa_h + \theta \kappa_{\mathrm{pat}}(C_{1/2}), \qquad 0 \le \theta \le 1. \tag{26}$$

Thus the discrete state-sum killing budget is fixed to roundoff while both splitting weights and conditional channel shapes may vary. M2D-F is not a continuation of the endpoint family M2D-E in the Supplemental Material: transport parameters, reaction radius, starts, and per-grid budget paths differ.

Sparse generator actions give f_t, f_{tt}, f_{ttt}, and an augmented exponential gives the parameter derivatives; the fold system is solved with its analytic 2×2 Jacobian at parameter tolerance 10^{-11}, under the same normalized acceptance guards as the encounter chain. The three

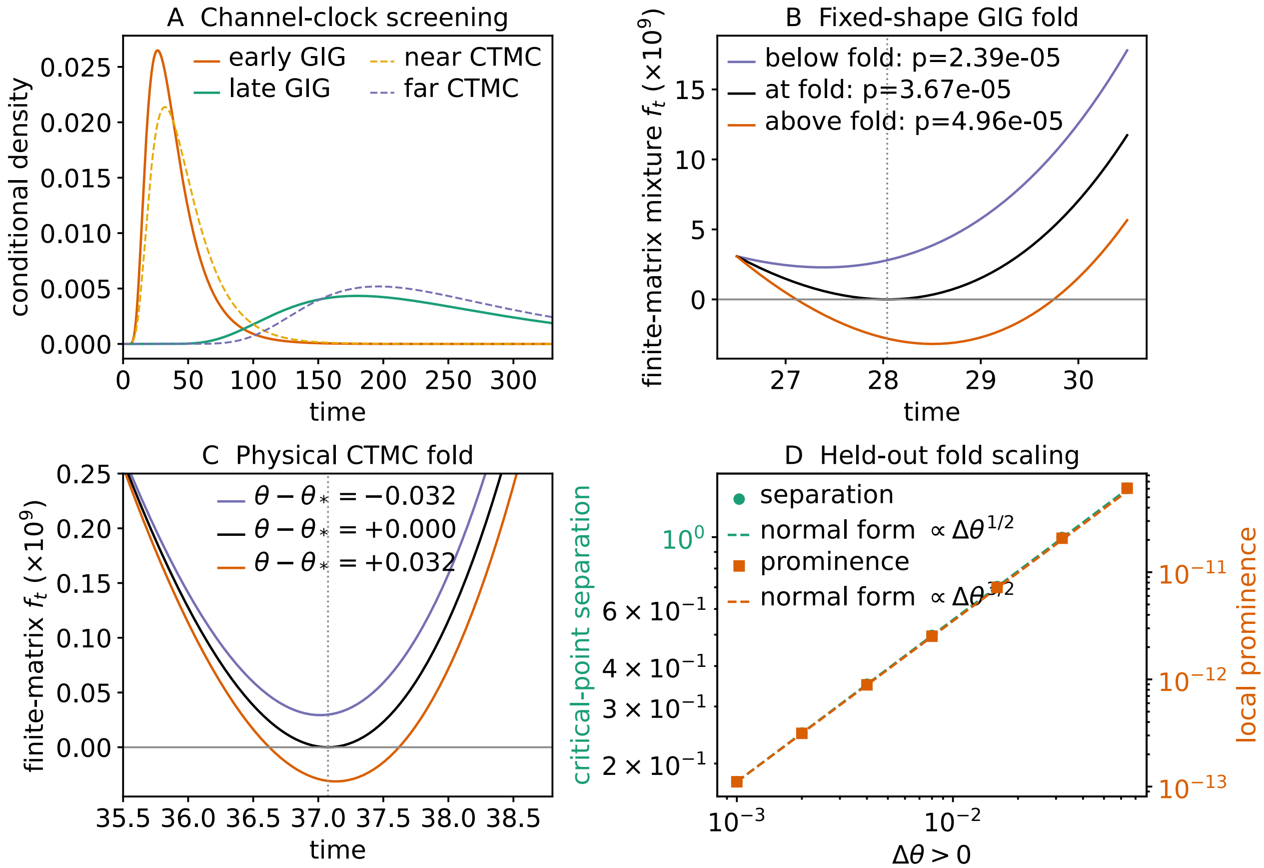

FIG. 2. Finite-CTMC fold certificate. The fold is located directly from matrix-exponential derivatives of the killed generator. Independent continuation recovers the 1/2 separation and 3/2 prominence exponents. The free-space GIG channel clocks shown in the figure are screening estimates, not replacements for the finite-matrix calculation; their derivation and clock-control audits are in the Supplemental Material [20]. The near-channel splitting probability at the fold is 8.95×10^{-6}, so the result establishes a mechanism rather than two comparably observable streams.

physical-interval folds are

$$\begin{aligned}(t_c,\theta_c)_{9\times5} &= (16.8093321, 0.27537300),\\ (t_c,\theta_c)_{11\times7} &= (18.0995323, 0.01381035),\\ (t_c,\theta_c)_{13\times9} &= (16.5807587, 0.25589201).\end{aligned} \tag{27}$$

The 9×5 residuals are below 3.5×10^{-17}, with $f_{ttt} = -1.34\times10^{-5}$ and $f_{t\theta} = 1.67\times10^{-4}$. The 11×7 residual pair is $(8.7\times10^{-19}, 3.5\times10^{-18})$, with $f_{ttt} = -7.81\times10^{-6}$ and $f_{t\theta} = 2.83\times10^{-4}$. The 13×9 residuals are below 2.8×10^{-17}, with $f_{ttt} = -5.96\times10^{-6}$ and $f_{t\theta} = 7.46\times10^{-5}$.

Independent continuation from eight control offsets $\mu = \theta-\theta_c \in [2\times10^{-4}, 0.05]$, fitted on the six offsets $\mu \le 0.01$, gives separation/prominence exponents $(0.4989, 1.5001)$, $(0.4933, 1.4843)$, and $(0.4993, 1.5039)$. All three generators therefore satisfy the local nondegenerate normal form. They do not form a convergent critical-control sequence: the state-count-matched θ_c values are nonmonotone and span 0.262. Replacing equal node weights in the budget by tensor-product boundary trapezoidal weights (per-axis weight $\frac{1}{2}$ on the two boundary nodes and 1 in the interior, tensorized over both particles), without changing the CTMC generators or masks, moves the folds to 0.6244, 0.2408, 0.4593, again nonmonotonically, with span 0.384. This one-factor audit shows that the finite-model fold is also sensitive to the budget measure.

Figure 3 collects the fixed-budget construction, the endpoint morphology audit, the derivative-located folds, and their independent normal-form scaling.

The two intervening grids are consistent with that caution. On 12×8, a nondegenerate continuation root lies at $(t_c,\theta_c) = (18.26877, -0.06158)$, outside the physical path. On 10×6, a curvature scan over $\theta\in[-0.2, 0.1]$, $t\in[0.01, 40]$ and four bounded least-squares starts (bounds $t\in[8,25]$, $\theta\in[-0.2,0]$) find no fold, with all starts reaching the same positive near miss $f_t = 3.8721\times10^{-6}$. This bounded search does not cover the interval $\theta\approx 0.25$–0.28 in which two of the three located folds sit; it is a bounded diagnostic, not evidence of absence. Detailed endpoint roots, budget quadrature, even-grid scans, and morphology settings are provided in the Supplemental Material [20].

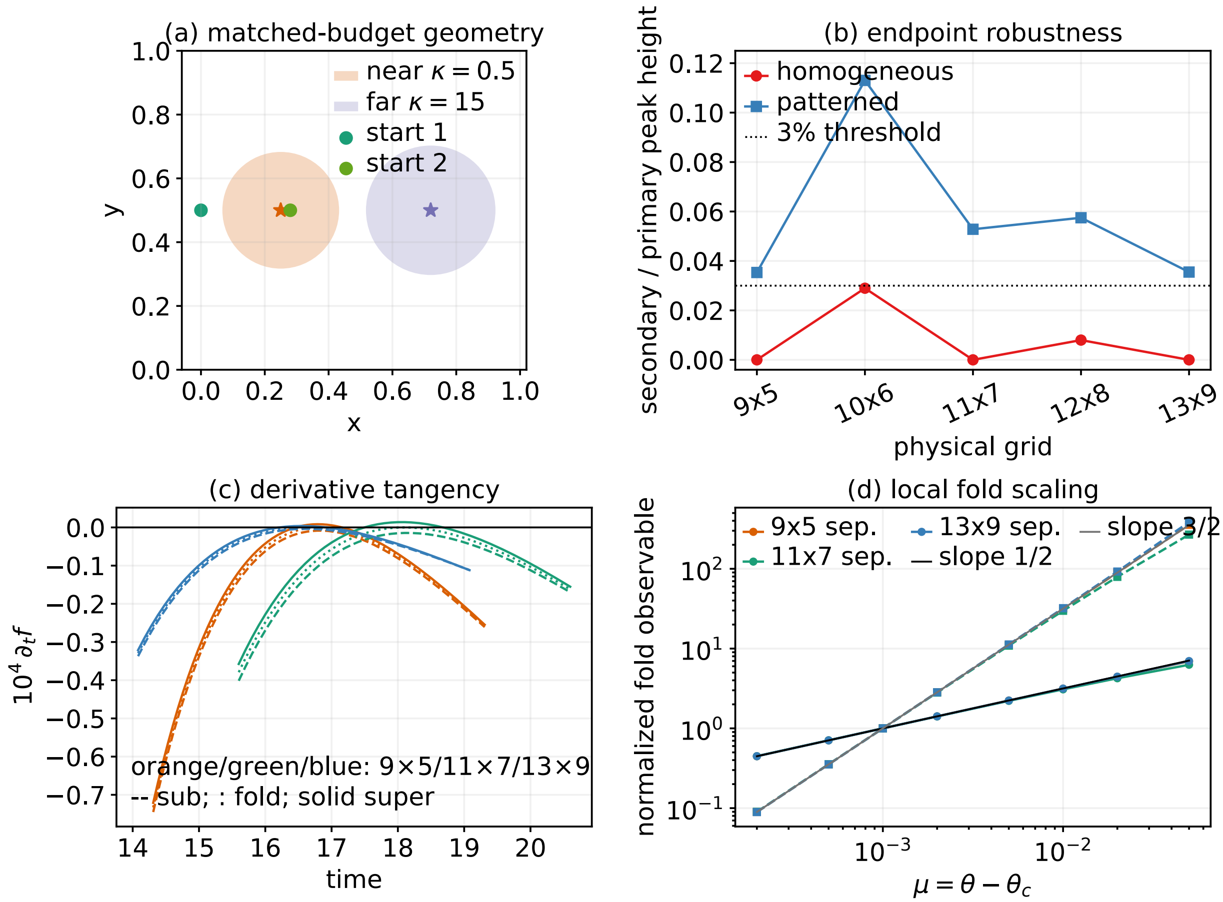

FIG. 3. Finite-radius two-dimensional matched-budget (M2D-F) fold certificate. (a) Spatial redistribution keeps the statewise killing sum fixed on each grid. (b) The endpoint audit distinguishes resolved secondary peaks from reported subthreshold ripples. (c) Analytic derivatives locate a tangency and its two-sided continuations on three finite grids. (d) Each grid recovers square-root separation and 3/2 prominence. The critical controls $0.275, 0.014, 0.256$ are explicitly nonmonotone and move under a second budget measure; they are not estimates of a continuum fold.

V. DISCUSSION AND SCOPE

The finite-model calculation separates three questions. Transport determines which temporal scales are available. The initial and reaction observables determine the signs and amplitudes of their spectral residues. Finally, a declared feasible reaction-field perturbation determines whether the critical-point equation can be crossed. Patch count alone addresses none of these questions: a support-rank-one counterexample can already be bimodal, whereas a residue sequence with three sign changes need not have three positive derivative roots.

The Green reduction and sensitivity calculus play complementary roles. $\mathcal{G}(s)$ reduces transport to the reaction support, while Eqs. (16)–(19) operate directly on the time-domain fold equation. A killed pole can be dark or cancel from a selected observable, and a density fold is not a spectral exceptional point. The spectral sign count is therefore used only to falsify an impossible finite reversible mode count, never to replace direct time-derivative root calculations.

The finite encounter CTMC confirms the derivative chain but has a very small minority channel. The M2D-F result more directly tests spatial redistribution at a fixed per-grid budget, yet its grid and budget-measure sensitivity prevents a continuum interpretation. The supporting Supplemental Material shows additionally that spatial patterning can promote an existing subthreshold maximum across an operational resolution threshold in a separate endpoint family, but is not necessary for bimodality; a uniformly reactive finite model provides a counterexample [20]. It also records catalytic-coordinate sensitivity and underresolved three-patch finite-grid examples. Those controls restrict causal language rather than broaden the principal claim.

The main unresolved step is a continuum bridge. Preservation of already simple modes needs derivative margins, whereas transfer of a nondegenerate fold needs convergence of

$$H_h = (\partial_t f_h, \partial_{tt} f_h)$$

and its time/control Jacobian near the critical point. Joint local C^3 convergence in time and control is sufficient. Mere C^2 convergence of the densities is not sufficient because third time derivatives can lose nondegeneracy. A cell-averaged, successively refined continuation with stable support quadrature, extrapolated critical control, and convergent parameter-Duhamel derivatives is therefore required before a continuum fold can be claimed.

Other limits remain explicit. The projected optimum is local and ignores finite-amplitude nonnegativity, box constraints, binary supports, and nonsmooth patch motion. The Green identity is a right-half-plane volume-reaction statement under its operator hypotheses; an unqualified Robin trace analogue and continuum meromorphic continuation are not supplied. The 2D grids use upwind transport and binary masks, and no independent Robin or Brownian solver validates the fold. The chosen discrete state-sum budget does not equalize pathwise hazards, splitting probabilities, or mean lifetimes. Capacity and GIG calculations in the Supplemental Material are calibration and screening layers, not centre-patterned modality theorems.

VI. CONCLUSIONS

For a declared finite encounter generator, the observable spectral residues give a necessary sign-variation gate, and the Fréchet–Duhamel derivative gives the exact local response of the critical-point equation to the killing field. Projection onto a budget tangent space then identifies the steepest unit feasible redistribution in a chosen metric. These statements link transport, reaction observability, and admissible control without equating the number of catalytic patches with the number of modes.

Two numerical certificates exercise the calculus. A finite two-reactant CTMC has a nondegenerate density fold with direct residual and transversality checks and the expected continuation exponents. A separate finite-radius M2D-F path has analogous folds on three finite grids under an exactly fixed per-grid killing sum. Their critical controls are nonmonotone and budget-measure sensitive, so the result stops at a finite-generator mechanism certificate. Establishing a continuum fold requires a new convergence calculation; it is not inferred from the present grids.

DATA AVAILABILITY

The source code, numerical tables, vector figures, and machine-readable provenance manifests that reproduce every reported number are available from the author upon reasonable request, and will accompany the published article as a public, versioned archive.

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